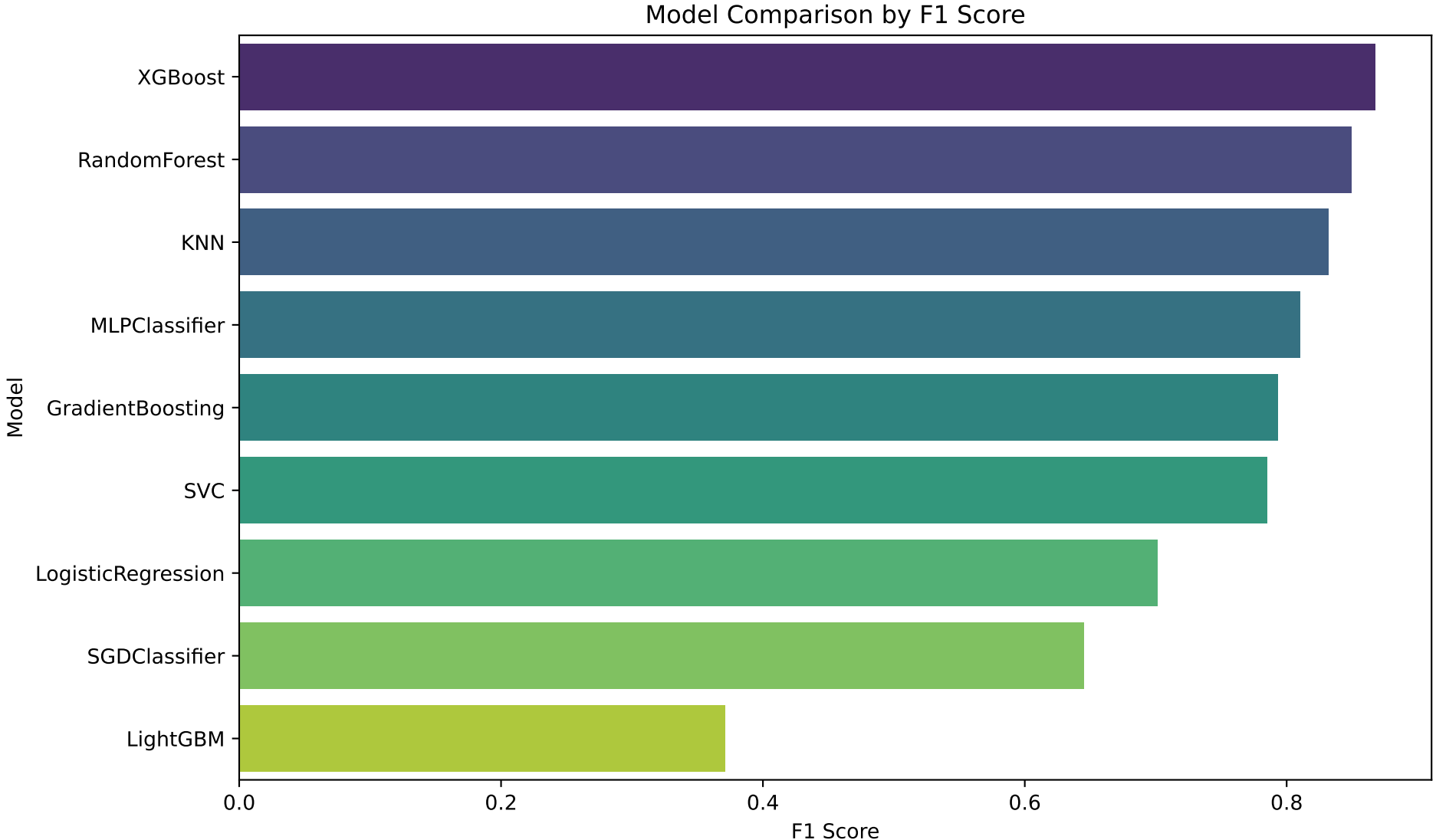
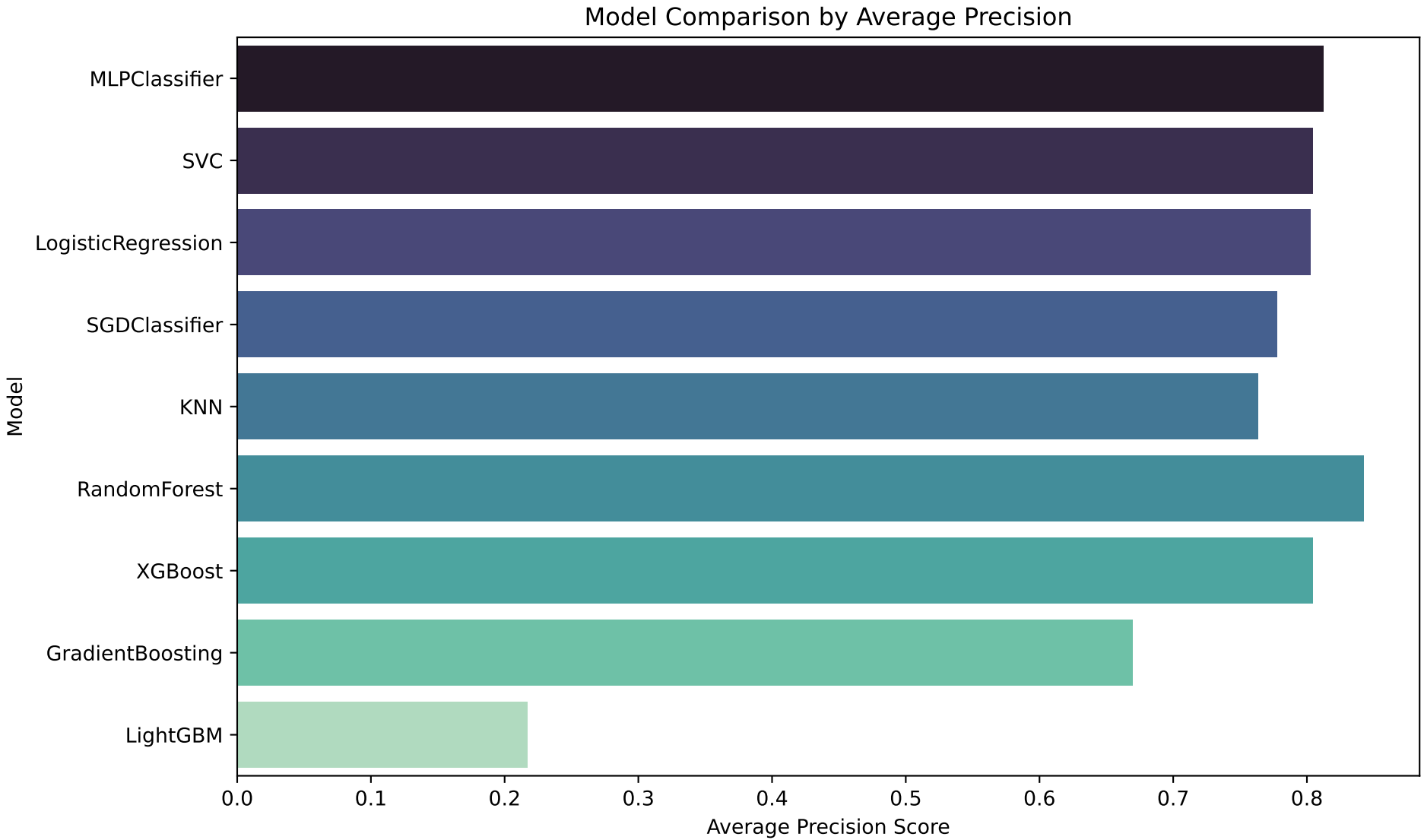
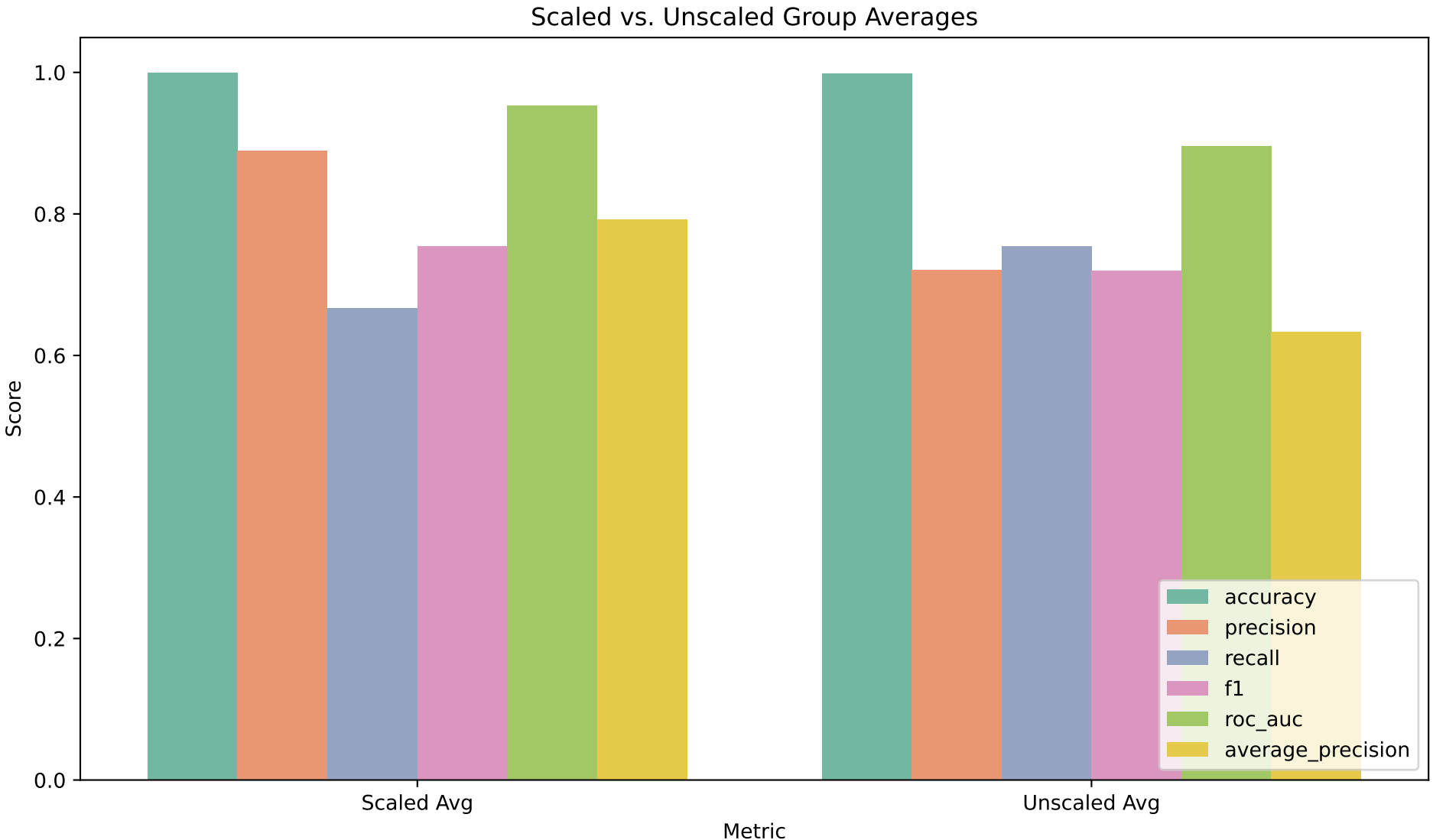
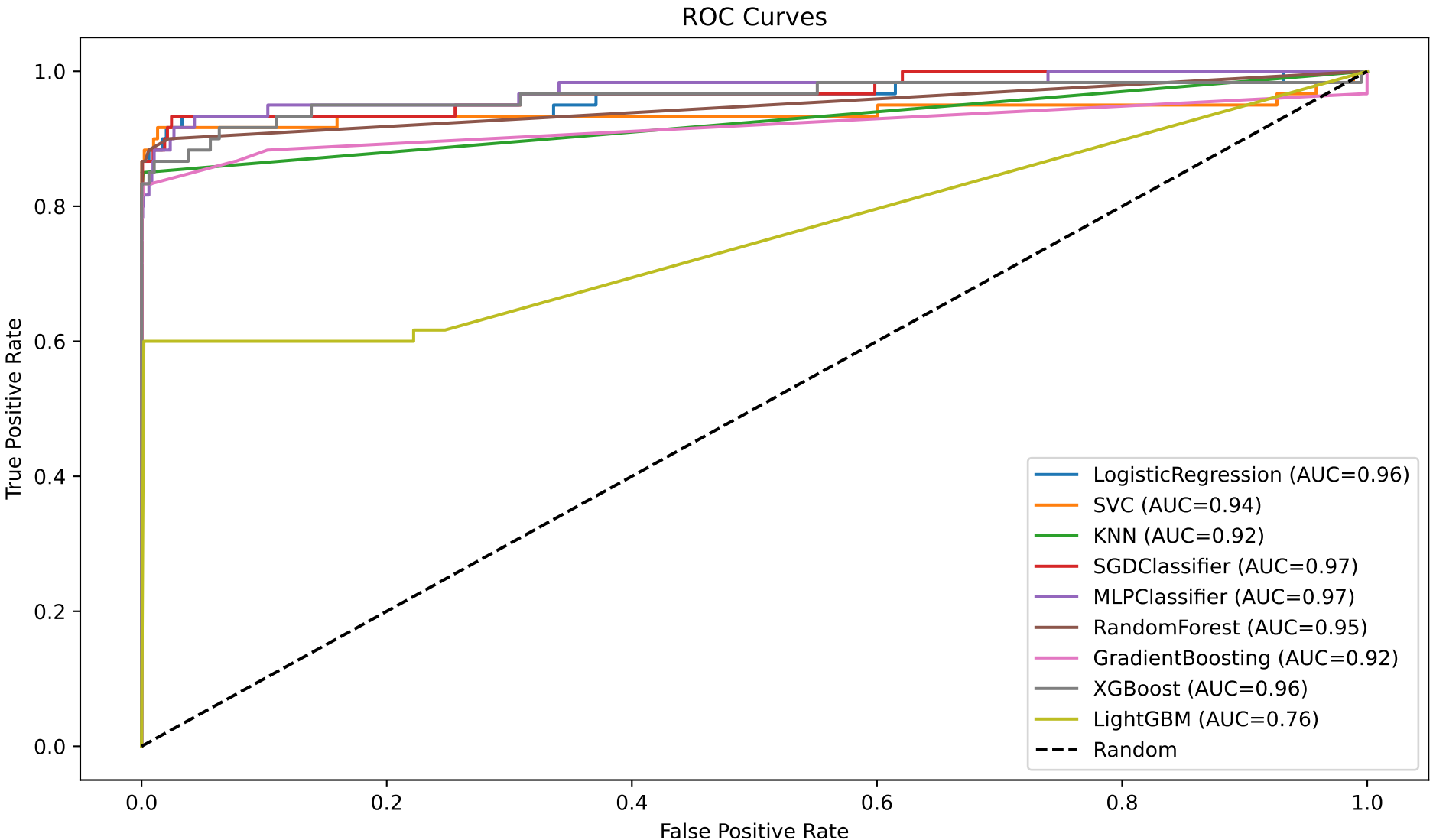
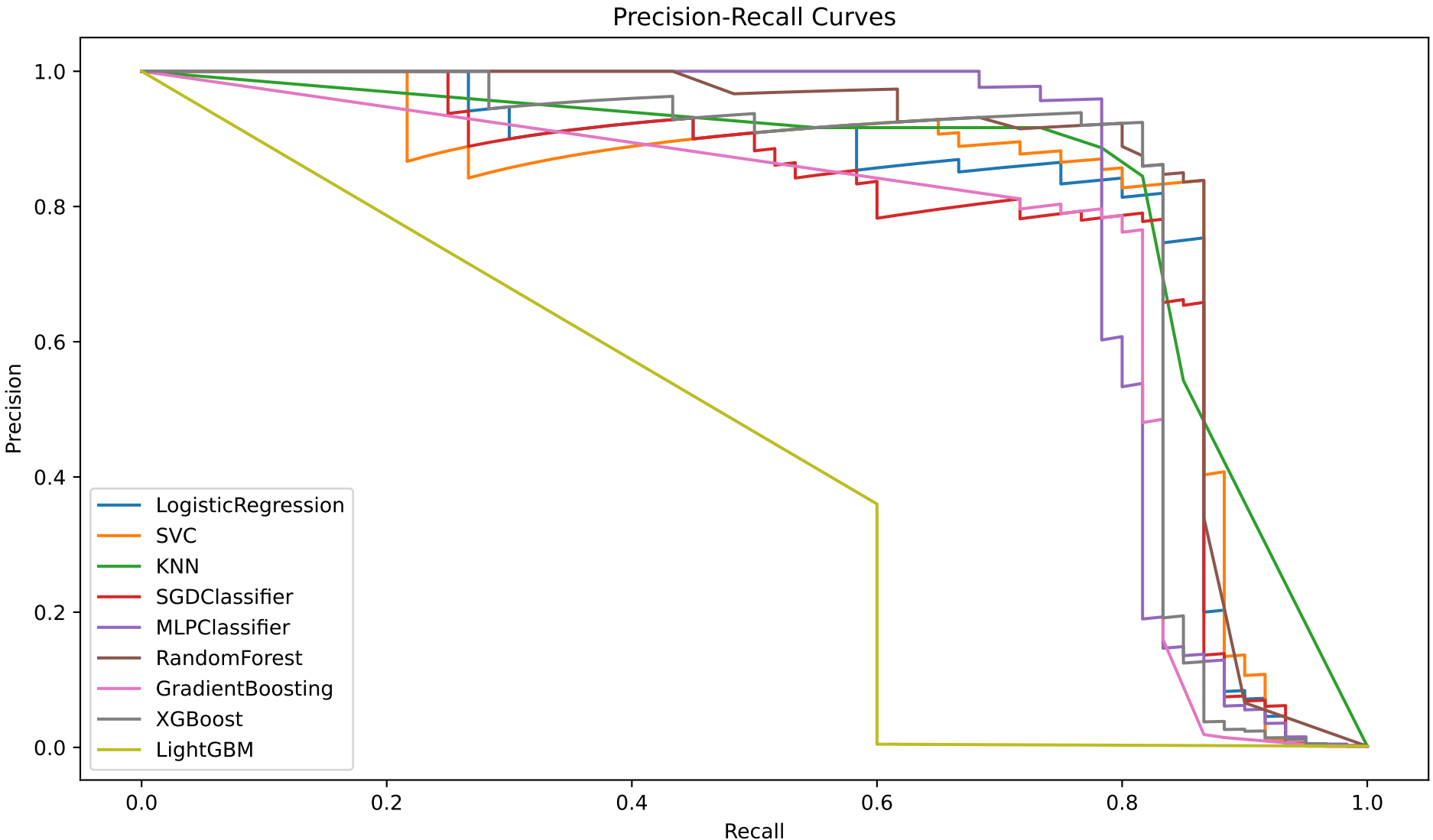
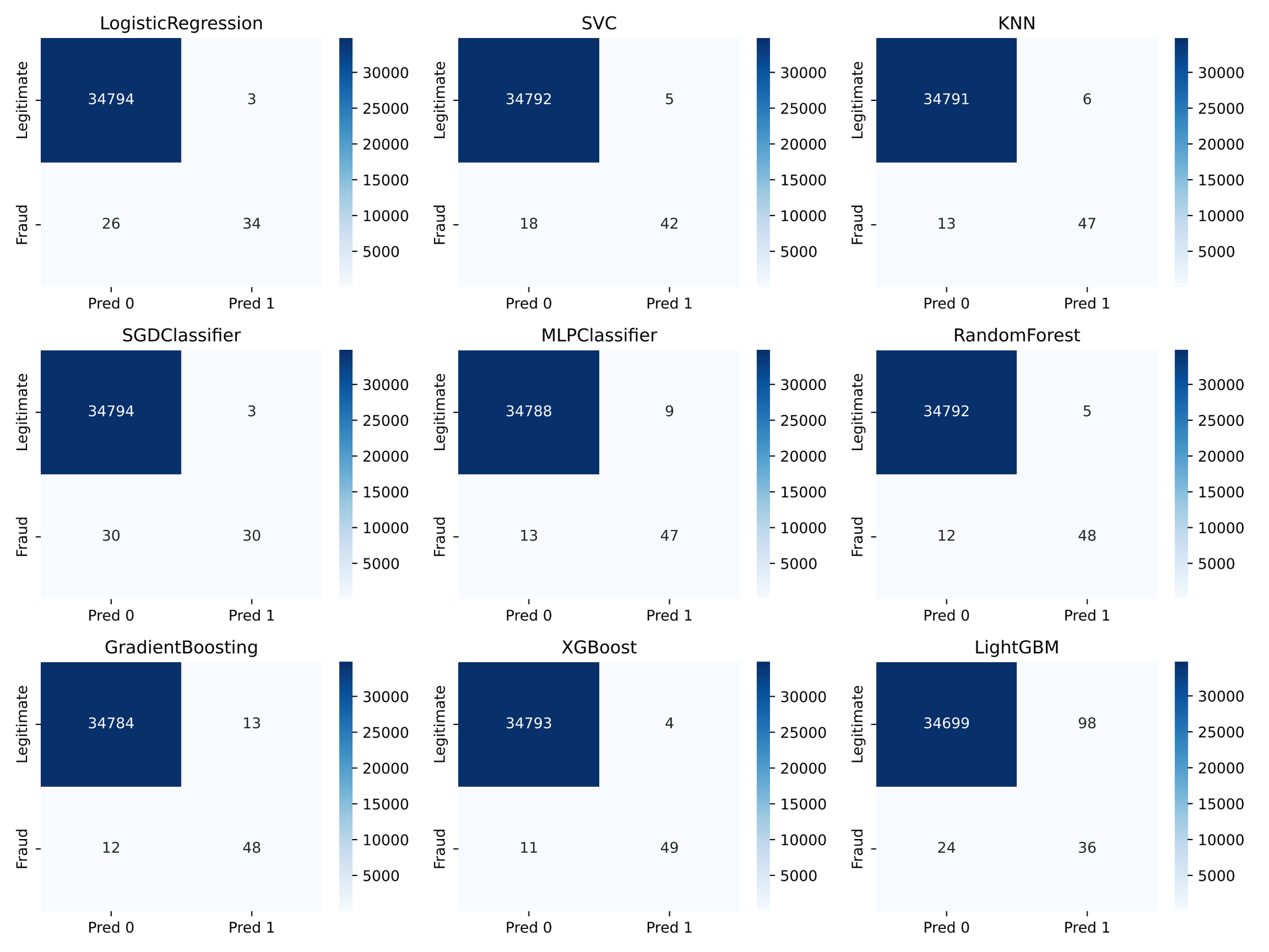

Model Comparison by F1 Score
XGBoost
RandomForest
KNN
MLPClassifier
GradientBoosting
SVC
LogisticRegression
SGDClassifier
LightGBM
Model
0.0
0.2
0.4
0.6
0.8
F1 Score


Model Comparison by Average Precision
MLPClassifier
SVC
LogisticRegression
SGDClassifier
KNN
RandomForest
XGBoost
GradientBoosting
LightGBM
Model
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Average Precision Score


Scaled vs. Unscaled Group Averages
1.0
0.8
0.6
0.4
0.2
0.0
Score
Scaled Avg
Unscaled Avg
Metric
accuracy
precision
recall
f1
roc_auc
average_precision


ROC Curves
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
LogisticRegression (AUC=0.96)
SVC (AUC=0.94)
KNN (AUC=0.92)
SGDClassifier (AUC=0.97)
MLPClassifier (AUC=0.97)
RandomForest (AUC=0.95)
GradientBoosting (AUC=0.92)
XGBoost (AUC=0.96)
LightGBM (AUC=0.76)
Random


Precision-Recall Curves
Precision
Recall
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
LogisticRegression
SVC
KNN
SGDClassifier
MLPClassifier
RandomForest
GradientBoosting
XGBoost
LightGBM


LogisticRegression
Legitimate
Fraud
34794
3
26
34
Pred 0
Pred 1
30000
25000
20000
15000
10000
5000
SVC
34792
5
18
42
KNN
34791
6
13
47
SGDClassifier
34794
3
30
30
MLPClassifier
34788
9
13
47
RandomForest
34792
5
12
48
GradientBoosting
34784
13
12
48
XGBoost
34793
4
11
49
LightGBM
34699
98
24
36